

SEBASTIAN CRUZ

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EDUCATION AND CERTIFICATIONS

Middlebury College | Middlebury, VT

Expected May 2027

Bachelor of Arts in Computer Science | GPA: 3.82/4.00

Relevant Coursework: Relevant Coursework: Software Development, AI, Algorithms, Machine Learning

Honors & Awards: College Scholar's List (Highest Academic Honors), QuestBridge Scholar

Certifications: CodePath Web Development (Intro & Intermediate), Google Project Management

EXPERIENCE

AI Lead and Backend Developer | Altheros Capital | Remote

Spring 2026 – Present

- Promoted to AI Lead after assuming technical leadership responsibilities for the backend AI team, coordinating development efforts, reviewing pull requests, assigning tasks, and contributing to sprint planning and architectural decisions.
- Designed and implemented Retrieval-Augmented Generation (RAG) infrastructure in Python, building document ingestion pipelines that process, chunk, embed, and store knowledge in PostgreSQL/pgvector to support AI-powered knowledge retrieval.
- Developed backend APIs, data models, and AI services using Python and Pydantic while researching and evaluating new technologies to improve system scalability, maintainability, and production AI workflows.

Backend Developer | Text Maddie | Remote

Summer 2026

- Developed backend workflows for AI-powered lead conversion agents, implementing Python logic that automated follow-ups, appointment scheduling, CRM updates, and integrations across HubSpot, Slack, and messaging platforms.
- Engineered AI agent behaviors, workflows, and tool integrations to support natural customer conversations while triggering automated business processes and coordinating multi-agent interactions.
- Built client onboarding and landing pages while investigating production issues through log analysis, workflow tracing, and backend debugging to ensure reliable AI agent performance.

Research Assistant | Middlebury College | Middlebury VT

Summer 2025

- Contributed to [NPSV-deep](#), a machine learning project for structural variant genotyping using deep neural networks in PyTorch.
- Implemented InfoNCE and RINCE loss functions, improving embedding-based classification accuracy by 0.5–2.9% across 4 datasets and increasing non-reference concordance by up to 3%.
- Trained and evaluated deep learning models in PyTorch Lightning using TensorBoard, optimizing parameters such as temperature scalars and rank thresholds, and using version control to maintain working iterations.

TECHNICAL PROJECTS

Scrum Master & Developer | [MiddBin](#) | Middlebury, VT

Fall 2025

- Developed a full-stack marketplace web application using Next.js, React, TypeScript, and Supabase, collaborated with a team through GitHub pull requests and served as Scrum Master.
- Designed and implemented a dynamic listing creation form with conditional inputs (categories, clothing sizes, gender, shoe sizing) and polished UI transitions using CSS to improve usability and data validation.
- Engineered a scalable image upload system using Supabase Storage, including multi-image previews, gallery views, and automatic cleanup on listing deletion and fully integrated it with listing grids and detail pages.

ADDITIONAL INFORMATION

Technical Skills: Python, FastAPI, Pydantic, PostgreSQL, React, [Next.js](#), TypeScript, JavaScript, PyTorch

Languages: English (Fluent), Spanish (Native), Portuguese (Intermediate)

Interests: Visual Arts, Traveling, Video Games, Films